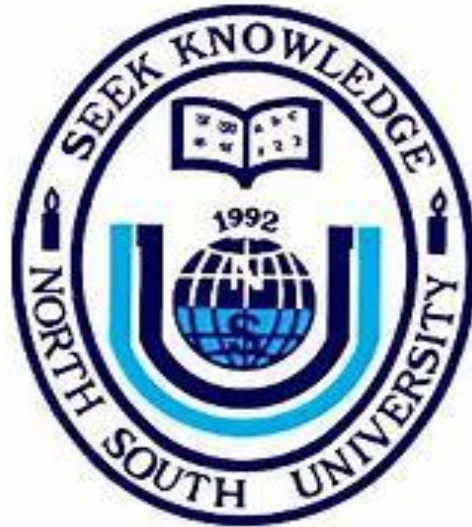


North South University
Department of Electrical & Computer Engineering
PROJECT REPORT



Project Title:

Inventory Management and Billing Software

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Introduction:

This is an Inventory Management and Billing software for a Vinyl and CD shop named “T1 Trade”. Through this software, a customer can buy games, music, movies of his choice very easily. Besides, T1 Trade has arranged discounts for customers.

Features:

The Inventory Management and Billing Software for the Vinyl & CD Shop encompasses a wide range of features, providing a comprehensive solution for product management, inventory tracking, billing, and discounts. This section delves into the key features of the software, organized into relevant categories.

Product Management

The software introduces a flexible and hierarchical product management system, accommodating various product categories such as music albums, movies, and games. The following features are highlighted:

Product Class Attributes:

The core of the product management system is the Product class, which encapsulates the essential attributes of each product, including its name, price, and category.

Hierarchy of Specialized Classes:

The software establishes a hierarchy of specialized classes that extends from the Product class, including:

StockableProduct: A class responsible for tracking the number of items stocked for each product.

Game, Music, and Movie: Product-specific classes that inherit the attributes and functionality of StockableProduct. These classes cater to the distinctive characteristics of games, music albums, and movies.

Inventory Management

The Inventory Management module ensures efficient stock tracking and organization, allowing the shop to maintain accurate inventory records.

Inventory Class:

The Inventory class serves as the repository for all stockable products. It offers a category-based management system, providing a dedicated space for each product category.

Stockable Products:

The software maintains records of each stockable product, capturing

crucial details such as name, price, available stock, and category. This ensures that the shop owner can readily access product information.

Billing and Discount

The billing component of the software simplifies the checkout process for customers and facilitates discounts based on predefined criteria:

Product Discounts:

The software introduces a dynamic system for applying product discounts. Each product may have its unique discount, enhancing the shop's ability to attract customers with varying preferences.

"Full House" Discount:

A special discount, known as the "full house" discount, is offered to customers who purchase at least one product from each category. This discount amounts to 50% off the total price and is automatically applied if eligible.

Sorting

Efficient inventory management is further enhanced by enabling sorting capabilities for the shop owner:

Sorting by Price:

The software allows the shop owner to sort the inventory by price, enabling quick identification of high-value or low-cost items.

Sorting by Available Stocks:

Inventory sorting can also be performed based on available stock levels. This feature aids in identifying products that require restocking or promotion.

Use Case Examples

To illustrate the practical application of the software, here are a few real-world use case examples:

Customer selects a music album, a movie, and a game, triggering the "full house" discount. Shop owner views the inventory, sorting items by price to identify premium vinyl records. A customer purchases multiple items, and the software calculates the total cost, including applicable product discount. These use cases demonstrate the software's effectiveness in handling day-to-day operations and enhancing the shopping experience for both customers and shop owners.

Methodology:

We analyzed the provided problem statement, identifying key features and constraints. Documented detailed requirements, including the product hierarchy, inventory management, billing system, and discount logic.

Class Hierarchy Design:

- Developed UML class diagrams based on the identified requirements.

- Designed abstract classes (Product), interfaces (Stockable, Iterable), and specialized classes (Game, Music, Movie).

Data Structure Design:

- We are using ArrayList as the primary data structure for the Inventory class.
- We implemented the Iterable interface for enhanced data structure iteration.

Object-Oriented Programming:

- We implemented classes and methods following the designed class hierarchy.
- It ensured proper encapsulation, inheritance, and polymorphism principles.
- We utilized the Iterator pattern for iterable collections.

Iterative Development:

- We adopted an incremental development approach, implementing and testing one module simultaneously.

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Price Sorting:

- We implemented the Comparator interface for sorting products by price.
- We validated the sorting functionality using test cases.

Stock Sorting:

- Similar implementation for sorting products by available stocks.
- We conducted tests to ensure accurate stock sorting.

Full House Discount Algorithm:

- We developed an algorithm to determine full-house discount eligibility.
- We integrated the discount logic into the billing system.

GUI(Graphical User Interface):

- Choose Window Builder as the GUI framework for its ease of use with Java Swing.
- Ensured compatibility with the existing codebase.
- Modified existing classes to accommodate GUI elements.
- Created new GUI-related classes and interfaces.
- Utilized Window Builder to design and implement various GUI components such as buttons, labels, text fields, and tables.
- Linked GUI components to corresponding backend functionalities.

- Implemented event handling for user interactions using ActionListeners and other relevant interfaces.
- Ensured seamless communication between the GUI and backend logic.
- Conducted integration testing to ensure seamless communication between the GUI and backend systems.
- Validated that user actions triggered the correct backend processes.
- Verified the accuracy of displayed data and real-time updates.

Conclusion:

In conclusion, the Inventory Management and Billing Software successfully addresses the specific needs outlined by the Vinyl & CD shop “T1 Trade”. The robust backend, intuitive user interface, and customer-centric pricing model position this software as a valuable tool for efficient inventory management and billing processes. The development team remains committed to ongoing support and continuous improvement to ensure the software's longevity and relevance in a dynamic business environment.